# No silver lining: Invasive goldfish in the Great Lakes

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> Seeing an orange-colored goldfish in your local lake is actually a sign of bad luck for native fish and plants.



This 14-inch goldfish was removed from the Niagara River by the U.S. Fish and Wildlife Service. Once goldfish are established in areas outside of their native range, they can significantly impact the ecosystem. Photo: USFWS

The common goldfish (*Carassius auratus*) is a freshwater fish native to Asia that is commonly kept as a pet, but which has become invasive in many areas of the United States. Common goldfish and other goldfish varieties are members of the family Cyprinidae, one of the most diverse fish families – with 11 recognized subfamilies that contain thousands of freshwater fish, including common carp (*Cyprinus carpio*). Wild goldfish typically have an olive green or silver color similar to other members of the Cyprinidae family, which may be surprising given the pet goldfishes' unique color and shine. The brightly-colored goldfish most are familiar with are the result of thousands of years of breeding in favor of the "fancy" orange color. Historically viewed as a sign of good luck and fortune, seeing a goldfish in your local lake is actually a sign of bad luck for native fish and plants.

## Goldfish life history

Contrary to popular belief, the size of a goldfish is not based on how big its tank is, but rather how much food it is able to find in its environment. Pet goldfish remain smaller in size, typically 1 to 2 inches, but wild fish average 14 inches in length. There are even reports of captured wild goldfish that have grown to 19 inches and weigh up to 9 pounds. In the wild, goldfish reproduction begins in the spring after temperatures have

risen. Recently hatched goldfish, referred to as fry, grow quickly in their first few weeks of life, and can live to be 10-15 years old. This is also true for pet goldfish – an important factor to consider when deciding whether to bring a goldfish home.

## Invasion biology

There are several aspects of goldfish biology that make them great invaders. Goldfish are hardy fish capable of tolerating a variety of temperatures. This means they can live in many areas outside of their native range. In addition, goldfish are unique in that they can survive in bodies of water that have anoxic conditions (little to no oxygen available). Research has shown that goldfish can survive without oxygen for three to five months, and are even able to survive in bodies of water that freeze solid during winter months. This enables goldfish to live in bodies of water that would be unsuitable for many other fish. Goldfish are also willing to eat almost anything. Their ability to consume a diverse array of resources increases the likelihood of them establishing themselves in a new environment.

## Impacts of invasion

Once goldfish are established in areas outside of their native range, they can significantly impact the ecosystem. The diet of goldfish includes snails, insects, fish eggs, and small fish. Their behavior and general diet make them both a competitor and predator of native species. Additionally, the foraging behavior of goldfish, commonly referred to as 'suck and spit' feeding, uproots aquatic vegetation, reduces water clarity, and releases nutrients that can promote algae growth. This particular feeding method can destroy aquatic vegetation and reduce the amount of sunlight entering the water, limiting the regrowth and establishment of uprooted vegetation. The decrease in water clarity can lower overall water quality and negatively impact local species that require sight to search for prey. The release of excess nutrients can lead to algal blooms that threaten animal welfare, human health, and water quality. Removing an established goldfish population is nearly impossible, and the economic costs are significant due to their abundance: for instance, a series of 10 goldfish management events in Minnesota removed an estimated 1.6 million goldfish from a single chain of lakes! The best way to protect against an invasion of goldfish is to prevent their introduction in the first place.

## How to help

First and foremost, do not release any unwanted goldfish into local waterways. The dumping of non-native fish is considered stocking, and is illegal in Michigan without a permit. While it may seem like a humane way to rehome your goldfish, there may be negative effects for the local environment. If you or someone you know has an unwanted goldfish, there are several responsible alternatives to releasing it locally:

- Return your fish to the pet retailer.
- Rehome your fish to someone else. There are many resources available to help rehome an unwanted pet, such as social media groups.
- Bring your pet to a humane society or attend a Habitattitude™ exotic pet surrender event.
- Donate your fish to a local school or education center.
- Advertise your fish through online forums dedicated to aquariums and fish hobbyists.
- Talk to your local veterinarian or pet retailer for humane disposal options.

Suspected goldfish sighting in your area? Report it to the Nonindigenous Aquatic Species (NAS) program through the Sighting Report Form.

## Additional resources

- Goldfish (*Carassius auratus*) Species Profile – An extensive summary of goldfish identification, ecology, and invasion in the U.S.
- Habitattitude™ U.S. – The national campaign addressing pet species as aquatic invasive species.
- Don't Let it Loose – A campaign promoting responsible pet ownership, including resources to help you rehome an unwanted pet, supported by the US Fish and Wildlife Service.
- Fish of the Week Podcast – A podcast produced by the Fish and Wildlife Service that is equal parts informative and entertaining. They detail the current status and impacts of goldfish in Season 2, Episode 50.
- Reduce Invasive Pet and Plant Escapes (RIPPLE) - Education initiatives coordinated by Michigan State University Extension, the program offers information to aquarium and water gardener professionals, retailers and hobbyists about what to do with unwanted plants and animals.

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*This article was prepared by Michigan Sea Grant under award NA18OAR4170102 from the National Oceanic and Atmospheric Administration, U.S. Department of Commerce through the Regents of the University of Michigan. The statement, findings, conclusions, and recommendations are those of the author(s) and do not necessarily reflect the views of the National Oceanic and Atmospheric Administration, the Department of Commerce, or the Regents of the University of Michigan.*

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